Junghye Kim

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github.com/junghye01



EDUCATION

M.S. in Artificial Intelligence, Korea University, Seoul, Korea (2025.09 - 2027.08)

Advisor : Sungwoong Kim, Sungbin Lim

Research Focus : Multimodal Learning, Video Understanding, and Visual Reasoning

B.S. in Dongguk University, Seoul, Korea (2020.02-2024.02)

Department of Information & Communication Engineering
Double major in DataScience software
Cumulative GPA: 4.26/4.5
Honors : Summa Cum Laude

RESEARCH INTERESTS

Multimodal Reasoning, Video Understanding, Human-Object Interaction, Temporal and Causal Reasoning

RESEARCH EXPERIENCE

Graduate Researcher, AGILAB, Korea University (2025.09-Present)

Developed a video reward model integrating spatio-temporal scene graph grounding with structured plan-and-verify reasoning for fine-grained text-video alignment evaluation. Contributed to method design, reasoning data construction, model training, and experimental evaluation, resulting in a co-first-author paper at EMNLP 2026 Main Conference.

AGILAB Intern, Korea University (2025.04-2025.08)

Conducted research on video reward modeling for capturing fine-grained temporal dynamics and efficient inference techniques for large language models.

Researcher, Seoul Asan Medical Center (2024.08-2025.04)

Conducted CT-based cancer diagnosis research by developing a 3D benign lymph node segmentation model and a weakly supervised liver tumor segmentation model using synthetic tumor generation.

Research Intern, MEDAI (2023.11-2023.12)

Conducted research on CT reconstruction and a self-supervised CT image interpolation framework using image masked modeling.

Researcher, Wbridge Engineering Research Program (2023.04-2023.10)

Developed an adaptive prompt-matching technique that dynamically applies prompts based on question types; resulted in a publication in KSC 2023.

Data Analysis & Machine Intelligence LAB Intern, Dongguk University (2023.01-2023.10)

Conducted machine learning and computer vision research on hierarchical network attack classification and long-tailed multi-label image classification.

Research Intern, NIA AI Data Construction Project (2022.08-2022.12)

Conducted research on 3D human action analysis by developing the DGU-HAO benchmark dataset and systematically evaluating its effectiveness using skeleton-based action recognition models; resulted in a publication in IEEE Access.

PUBLICATIONS

Plan-and-Verify Video Reward Reasoning with Spatio-Temporal Scene Graph Grounding
Hyomin Kim*, **Junghye Kim***, Joanie Hayoun Chung, Yoonjin Oh, Kyungjae Lee, Sungbin Lim, Sungwoong Kim.
EMNLP 2026 — Main Conference (* Equal contribution)

DGU-HAO : A Dataset with Daily Life Objects for Comprehensive 3D Human Action Analysis
Jiho Park, **Junghye Kim**, Yujung Gil, Dongho Kim

IEEE ACCESS, 2024

APMT : Adaptive Prompt-Matching Technique to enhance your answers
Yunji Kim, **Junghye Kim***, Yeosol Heo*, Jihie Kim (* : Equal Contribution)
Korea Software Congress, 2023

PATENTS

Apparatus and Method of Classifying and Detecting Real-Time Abnormal Behavior Based on Artificial Intelligence
(No. 10-0004987(2024)) (Submitted) Sangsoo Lim, **Junghye Kim**, Junho Choi, Seunghyeon Jung

CERTIFICATIONS

Advanced Data Analytics Semi-Professional(ADsP) Certification FEB 2022

TEACHING EXPERIENCE

'Machine Learning and Data Science' - Teaching Assistant SEP 2023- DEC 2023
- Led practical sessions applying machine learning algorithms to real-world datasets

Wbridge, Korean Women in Science and Technology Foundation - Mentor APRIL 2023- OCT 2023
- Mentored high school students on open-source project development and collaboration using Git and GitHub
- Guided students in research team participation and study engagement

'Open source Software Practice' - Teaching Assistant MAR 2023- JUNE 2023
- Led practical sessions on Docker-based deployment of Flask web applications
- Mentored 50 students through Q&A and interactive discussions

HONORS AND AWARDS

2024 Dongguk University Summa cum laude
2022 Dongguk University Academic Excellence Scholarship
2021 Top Student Scholarship by Course ('General Physics and Experiment 2')

LANGUAGE PROFICIENCY

Fluent in **English(TOEIC 905)** and Native in **Korean**

TECHNICAL SKILLS

Languages: Python, C, Javascript
Tools : Python, C, JavaScript, PyTorch, Hugging Face Transformers, OpenCV, NumPy, Git, GitHub, Docker, AWS, Jenkins

PUBLICATIONS

EMNLP (2026) - Main Conference	
Title	Plan-and-Verify Video Reward Reasoning with Spatio-Temporal Scene Graph Grounding
Author	Hyomin Kim*, Junghye Kim*, Joanie Hayoun Chung, Yoonjin Oh, Kyungjae Lee, Sungbin Lim, Sungwoong Kim
Keywords	Text-to-Video Evaluation, Video Reward Model, Semantic Alignment, Spatio-Temporal Scene Graph, Video Reasoning, Multimodal Large Language Model.
Role	Research Design, Method Development, Data Construction, Model Training, Experiments and Analysis, Paper Writing and Submission

Background and Objective:Recent text-to-video (T2V) generation models have rapidly improved in visual quality, increasing the need for reliable reward models that can evaluate whether generated videos faithfully satisfy complex text prompts. However, existing reasoning-based video reward models often rely on free-form reasoning, making it difficult to ensure that every prompt requirement is explicitly verified and that each judgment is grounded in concrete visual evidence. To address these limitations, we propose SG-PVR, a video reward model that combines structured plan-and-verify reasoning with spatio-temporal scene graph grounding for fine-grained text-video semantic alignment evaluation.

Methods: SG-PVR first extracts a spatio-temporal scene graph from the input video, representing entities, attributes, relations, and their temporal information. The model then decomposes the text prompt into atomic verification claims through a verification planning stage. Each claim is independently verified using both the raw video and the structured scene graph as visual evidence, and the resulting claim-level judgments are aggregated into final semantic-alignment and video-quality scores. The model is based on Qwen3.5-9B and is trained through two-stage supervised fine-tuning: scene graph generation using approximately 307K refined training pairs, followed by reasoning training using approximately 83K constructed reasoning traces.

Results and Conclusion:SG-PVR demonstrated strong performance across multiple video reward benchmarks, particularly in fine-grained temporal reasoning. On a 600-pair temporal semantics test set, it achieved 31.4% overall accuracy. Ablation studies showed that removing the scene graph reduced performance to 30.5%, while replacing plan-and-verify reasoning with free-form reasoning further decreased it to 21.8%, highlighting the benefit of structured grounding and explicit verification.

As a test-time reranker for T2V generation, SG-PVR also improved compositional alignment, outperforming Random and VideoScore2 across all evaluated T2V-CompBench dimensions and surpassing OmniScore on five of seven dimensions. Overall, the results show that decomposing prompt requirements and grounding each verification step in spatio-temporal evidence enables more reliable evaluation of complex text-video alignment.



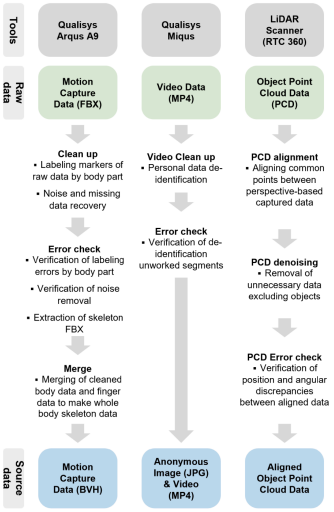
SG-PVR framework for structured text-video alignment evaluation via scene graph grounding and plan-and-verify reasoning.

IEEE ACCESS(2024)	
Title	DGU-HAO : A Dataset with Daily Life Objects for Comprehensive 3D Human Action Analysis
Author	Jiho Park, Junghye Kim, Yujung Gil, Dongho Kim
Keywords	3D human action analysis, human action recognition, human activity understanding, motion capture, multi-modal dataset.
Role	Data Preprocessing, Model Training, Thesis Writing and Submission

Background and Objective:Existing 3D human action datasets often lack detailed object interactions, limiting their real-world applicability. To address this issue, DGU-HAO introduces a dataset that incorporates 63 action classes interacting with 60 common furniture and electronic devices. This dataset integrates RGB videos, 3D skeleton data, and point cloud (PCD) data into a single JSON file, significantly reducing preprocessing efforts and making it more accessible for researchers.

Methods:DGU-HAO was built using 67,505 motion capture samples collected from 126 subjects. Each action class consists of three distinct phases (approach → interaction → retreat), and the 3D skeleton data includes precise coordinates for 25 joints. During preprocessing, BVH files were converted into CSV and JSON formats, ensuring a unified global coordinate system for enhanced usability.
For dataset validation, the MMNet model was trained on 80% of the data and tested on the remaining 20%, using both skeletal joints and bone connections as input, thereby improving motion representation and recognition accuracy.

Results and Conclusion:The MMNet model achieved Top-1 classification accuracy of 91.51% using the Skeleton Joint method and 92.29% using the Skeleton Bone method, outperforming or matching existing 3D action datasets.
Additionally, the lowest classification accuracy across all 63 action classes remained above 98.67%, demonstrating the dataset's robustness and reliability.
Beyond action recognition, DGU-HAO is highly applicable for various research areas, including human-object interaction analysis, action generation, and real-time detection.



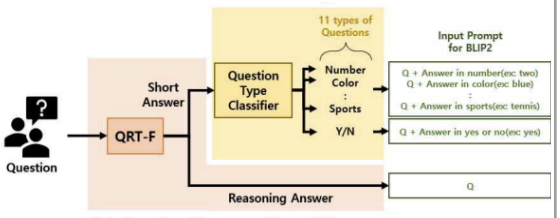
Comprehensive Setup for Capturing Multi-Modal Human Action Data in DGU-HAO Dataset

Korea Software Congress (2023)	
Title	APMT : Adaptive Prompt-Matching Technique to enhance your answers
Author	Yunji Kim, Junghye Kim*, Yeosol Heo*, Jihie Kim (* : Equal Contribution)
Keywords	Visual Question Answering, Vision-Language Model, Large Language Model, Prompt Engineering
Role	Experimental Design, Model Development, Prompt Application Experiments, Thesis Writing and Submission

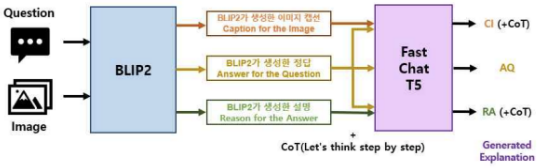
Background and Objective: Visual Question Answering (VQA) requires models to process both visual and textual information, but optimizing for both short factual answers and complex reasoning remains challenging. Existing approaches struggle to balance accuracy across different question types. To address this, we propose the Adaptive Prompt-Matching Technique (APMT), which dynamically adjusts prompts based on question type. APMT leverages Vision-Language models (BLIP2) for short-answer questions and refines descriptive responses using a Large Language Model (FastChat-T5) to enhance logical consistency and contextual awareness.

Methods:APMT classifies questions through a two-step process. The Question Response Type-Filter (QRT-F) distinguishes between short-answer and descriptive-answer types. Short-answer questions, such as yes/no, color, and counting, are further categorized by the Question Type Classifier (QTC) for optimized response generation using structured prompts in BLIP2. Descriptive questions are initially processed by BLIP2 and then refined by FastChat-T5 for improved clarity and coherence. To further enhance response quality, APMT employs adaptive prompt engineering. Short-answer questions are structured to maximize factual accuracy, while descriptive-answer questions incorporate image-caption-based prompting (CI, CI+CoT) to strengthen reasoning and coherence.

Results and Conclusion: Evaluations on two benchmark datasets demonstrate APMT's effectiveness. On the TDIUC dataset, APMT significantly improved short-answer accuracy, with BLEU scores increasing by a factor of 60. For descriptive-answer questions on the VCR dataset, image-caption-based prompts (CI, CI+CoT) yielded the highest BLEU, METEOR, ROUGE, and CIDEr scores, enhancing explanation quality. These results confirm that APMT improves both factual precision and reasoning-driven answers, leading to overall better performance in VQA tasks.



Question Processing with APMT



Explanation Generation through APMT

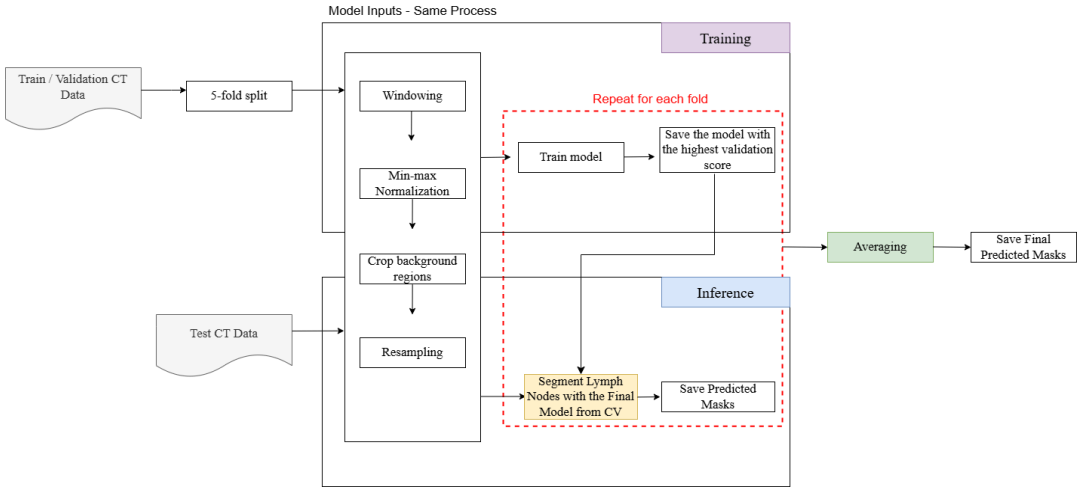
SELECTED RESEARCH PROJECTS

Title	A Dual-Resolution Mamba-based Approach to Mediastinal Lymph Nodes Segmentation in CT with weighted loss for class-imbalanced segmentation
Keywords	3D segmentation , Mediastinal Lymph nodes, CT, Mamba, nnU-Net, Multi-resolution, k-fold cross validation
Role	Experimental Design, Model Development and Testing

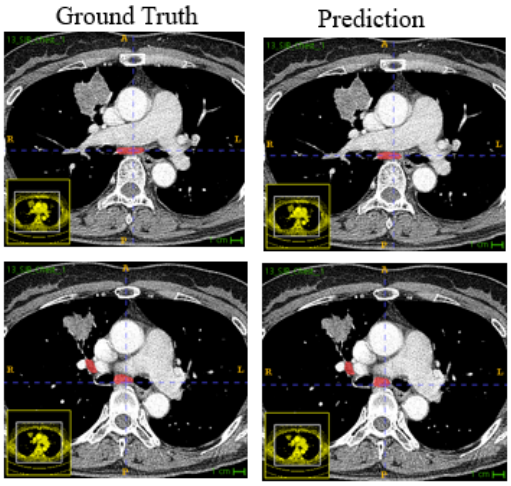
Background and Objective: Lymph node segmentation is vital for staging and treatment planning but remains challenging due to low contrast, size variability, and complex anatomical distribution. This study develops a high-performance model that enhances multi-scale feature extraction, refines boundary segmentation despite contrast limitations, and efficiently captures long-range dependencies in 3D CT scans while maintaining computational efficiency.

Methods: Lymph node segmentation remains challenging due to significant size and contrast variations, limiting the effectiveness of conventional architectures. CNNs, constrained by fixed receptive fields, struggle with long-range dependencies, while Transformers incur prohibitive computational costs for large 3D medical images. To address these issues, we adopt U-Mamba, a state-space model (SSM)-enhanced convolutional framework that efficiently processes long sequences while preserving local spatial structures. This architecture maintains scalability in volumetric data while mitigating the computational inefficiencies of self-attention mechanisms. A multi-resolution approach is employed, leveraging low-resolution 3D CT scans for coarse localization and high-resolution inputs for precise boundary refinement. Segmentation maps from both scales undergo connected-component analysis, filtering out false positives and enforcing structural consistency. To further improve specificity, foreground regions below 10,000 mm³ are discarded as they are prone to noise and over-segmentation artifacts. For training, we optimize a composite loss function—a weighted sum of Dice, cross-entropy, and Tversky losses—with a strong penalty on false negatives to improve recall, particularly for small or low-contrast lymph nodes.

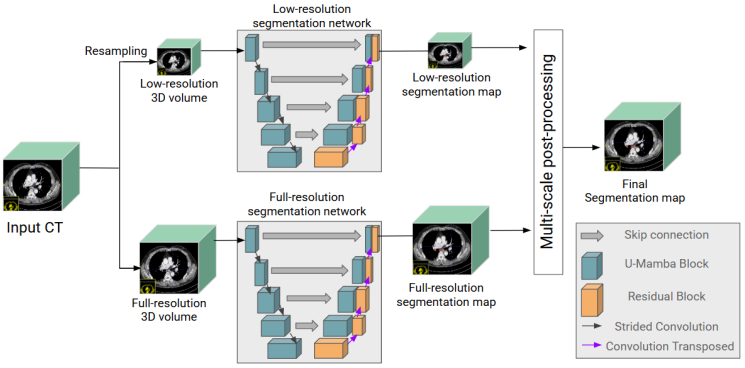
Results and Conclusion: The proposed dual-resolution U-Mamba framework achieved a patient-level sensitivity of 68.89%, positive predictive value (PPV) of 67.98%, and Dice similarity coefficient (DSC) of 0.6673 on the CT test set. Size-stratified evaluation demonstrated stronger performance for larger lymph nodes, achieving a detection ratio of 94.07% for nodes with a short-axis diameter ≥10 mm. Segmentation accuracy also increased with lymph node size, with DSC improving from 0.38 for nodes <6 mm to 0.70 for nodes ≥15 mm. These results demonstrate the framework's capability to detect and segment mediastinal lymph nodes in 3D CT, particularly for clinically significant larger nodes.



Model Training and Testing Process



Qualitative results during the test phase



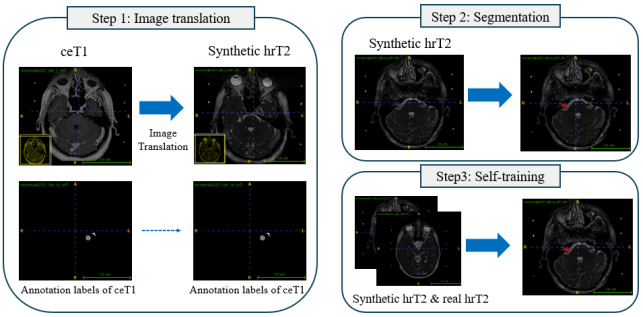
Overview of our multi-scale prediction fusion network Model Architecture

Title	Cross-Modality Domain Adaptation for Medical Image Segmentation
Keywords	Unsupervised Domain Adaptation, Medical Image Segmentation, Brain MRI
Role	Experimental Design, Model Development and Testing

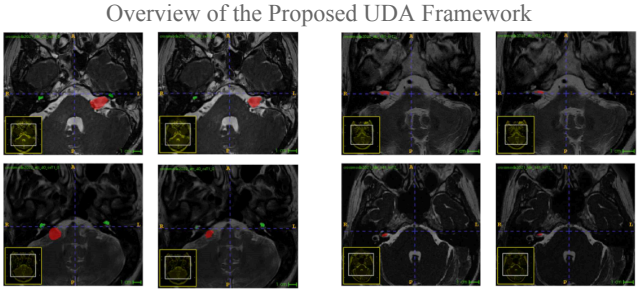
Background and objective: Unsupervised domain adaptation (UDA) is crucial for addressing performance drops with unseen modalities in medical image segmentation. While ceT1 scans offer high accuracy, they rely on costly gadolinium agents with safety risks. hrT2 scans are safer and cost-effective but lack sufficient contrast for accurate segmentation. This study addresses these challenges by using unpaired ceT1-hrT2 data with domain translation and self-training to enable accurate 3D segmentation of vestibular schwannoma (VS) and cochlea.

Methods: We propose a target-aware domain translation network using QSAttn to convert T1 scans into pseudo T2 images while preserving structures like VS and cochlea. A teacher-student self-training scheme bridges the gap between pseudo and real T2 data by leveraging T1 annotations to refine segmentation performance on T2 images.

Results and Conclusion: QSAttn with a segmentation decoder achieved Dice scores of 66.45% for VS and 68.75% for cochlea using pseudo T2 scans, while combining pseudo and real T2 data revealed domain gaps, reducing scores (VS: 58.13%, cochlea: 40.95%). This study highlights the effectiveness of target-aware translation and self-training but underscores the need for enhanced 3D translation networks and extended self-training to improve segmentation accuracy and clinical utility.



Overview of the Proposed UDA Framework



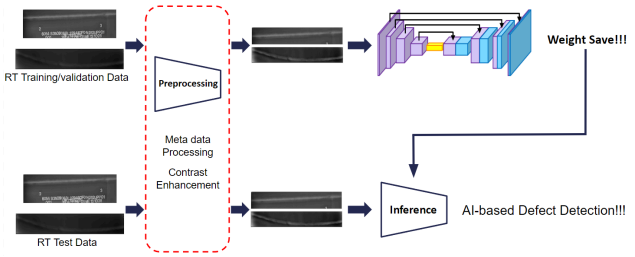
Qualitative test results trained on both pseudo T2 and real T2 data(Left : Ground Truth, Right : Prediction)

Title	AI Model Development for Welding Defect Detection
Keywords	Automated Welding defect detection, Radiographic images, Object Detection
Role	Experimental Design, Data Preprocessing, Model Development and Testing

Background and objective: Conventional radiography testing (RT) is slow, costly, and relies on manual interpretation. This study proposes an AI-driven Digital Radiography Test (DRT) to streamline workflows, reduce costs, and improve defect detection and classification efficiency.

Methods: We classify four defect types (Normal, Others, Porosity, Slag) using EasyOCR for metadata removal and CLAHE for contrast enhancement. Detection models were trained with data augmentation and evaluated by F1-scores. Non-max suppression refined bounding boxes, and weighted scores prioritized Porosity and Slag for model selection.

Results and Conclusion: Using test sets, Our model achieved the highest F1-scores for Porosity (78.2%), Slag (89%), and overall (89%), with a final weighted score of 84.68. These improvements, attributed to CLAHE and non-max suppression, enhanced defect visibility and reduced redundancy. AI-driven DRT offers faster, more cost-effective defect detection with superior performance and streamlined workflows.



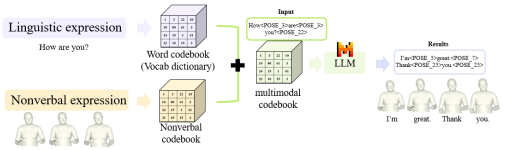
Model Development Strategy

Title	Development of a Multimodal Conversational Model
Keywords	Conversational AI, Human-Agent Interaction, Multimodal Learning
Role	Experimental Design, Data Preprocessing and Analysis of Collected Data, Model Training and Testing

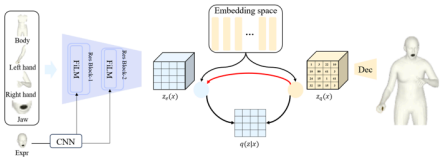
Background and objective: Nonverbal expressions like gestures and facial cues enrich communication but are often overlooked in text-only conversational AI, limiting natural interaction and engagement. To address this, we propose a multimodal conversation agent that integrates speech recognition and facial expression analysis to generate gestures, facial responses, and dialogue.

Methods: Using 10,000 hours of dual-speaker YouTube videos, we extract word, sentence, and utterance-level timestamps with whisperx and capture 3D nonverbal cues (e.g., gestures, facial expressions) via OSX. These are quantized into a unified multimodal codebook using VQ-VAE with FiLM conditioning layers, merging verbal and nonverbal inputs. The LLM (Mistral 7B with QLoRA) is trained on combined cues using reinforcement learning.

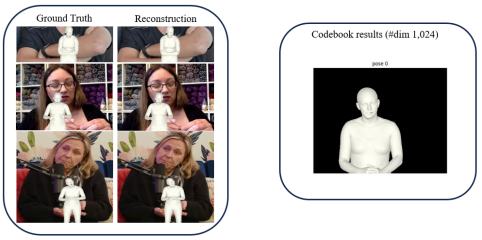
Results and Conclusion: Our multimodal agent processes 10,000 hours of video, generating 60,000 segments and 1.7M utterances with 3D nonverbal parameters. Using VQ-VAE with FiLM-based convolution and a joint loss setup, the model achieves superior reconstruction and perplexity. By appending quantized nonverbal tokens to text, it effectively unifies linguistic and nonverbal inputs, advancing conversational AI beyond text-only systems.



Multimodal Data Processing in Conversational AI



VQ-VAE Training Workflow



VQ-VAE Qualitative Results (Left : Reconstruction , Right: Codebook)